

Score-Identified Distributional Dynamics on *C. elegans*

A conditional-law functional connectome, a biologically-principled lag-resolved analysis, and the controls that turned a “significant neuromodulator signature” into a well-characterised, global-state-entangled lead

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Abstract

Conventional functional-connectivity and causal-discovery methods estimate a *conditional mean*: how one neuron’s activity predicts another’s future *level*. They are, by construction, blind to *distributional* interactions — how one neuron reshapes the *variance*, gain, or tails of another’s predictive law — which is precisely the regime in which slow, metabotropic and peptidergic neuromodulation is thought to operate. We develop **score-identified dynamics** (SID): estimate the full one-step conditional predictive density $\rho(y \mid H_t)$ and read out directed *mean*, *gain* ($\partial \log \text{Var}$), and *tail* channels as derivative-free covariances against the conditional score. We instantiate two estimators of the *same* object — a convex, closed-form Hyvärinen score matcher and a neural denoising-score-matching (DSM) model (the lineage of the prior “SBTG” network) — and validate the readout on a synthetic hidden-thermostat system, where the gain channel recovers a conditional-variance memory kernel that is provably invisible to every conditional-mean method (kernel correlation 0.998; mean-model $R^2 \approx 0$). Applying SID to whole-brain NeuroPAL calcium imaging (after auditing the prior pipeline for 29 bugs and fixing a connectome-orientation error that had invalidated its numbers), we find the closed-form SID mean channel is the equal-best single method across an ensemble of noisy anatomical and receptor-expression targets, and a maximum-likelihood non-Gaussian comparator (MDN) delivers the single strongest result where theory predicts (metabotropic serotonin, AUROC 0.78). We then pursue a *lag-resolved* signature of slow neuromodulation — and this is where the paper’s real contribution lies, because we *subject that one finding to a deliberate, escalating battery of controls*. An initial statistic (the log-slope of correspondence-vs-lag) suggested the peptidergic network’s distributional channel leans to longer lags than its mean channel; we critique that statistic, then test the effect across conditioning schemes (it is *partial*-specific), worm counts (*power*-limited), a coupling-destroying surrogate null (*real* above chance, $p = 0.008$), and an independent estimator (*not* reproduced). We redesign the metric from the biophysics — a per-connection expected-timescale, peakedness-aware *band concordance* with a full control battery, built and adversarially reviewed — and find that the peptidergic effect passes the surrogate, source-variance, and structural-dissociation controls, but **collapses when the shared global brain-state mode is removed** ($\Delta \text{BC} +3.4 \rightarrow -0.5$; $p 0.010 \rightarrow 0.69$), and is **not reproduced by a null-contrast-tuned neural DSM estimator** ($p = 0.88$). The honest conclusion: the lag signature is specific to the closed-form SID gain operator and entangled with global brain state — a well-characterised lead, not a robust directed-coupling result. What survives is a principled method, a decisive synthetic validation, a corrected pipeline, an overall-competitive real-data estimator, and a case study in controls-first neuroscience.

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1 Problem statement

The *C. elegans* nervous system runs two signalling systems on the same ~ 302 neurons. A **fast, wired** system — chemical synapses and gap junctions, the “connectome” — couples neurons over milliseconds: when neuron *A* fires it pushes neuron *B*’s activity up or down almost immediately. A **slow, largely extrasynaptic** system — monoamines (dopamine, serotonin, tyramine, octopamine)

and neuropeptides — diffuses to distant targets that express cognate G-protein-coupled receptors (GPCRs) and acts over seconds. Crucially, these two systems are expected to leave *different statistical footprints*. The wired system changes a target’s *expected next value* — a conditional-mean effect. Neuromodulation is expected to act on *excitability and gain*: a modulator does not so much set a target’s next value as reshape the *distribution* of what the target can do next — its variance, its propensity to burst, the heaviness of its tails — often *without* changing the mean.

Every workhorse method for inferring interactions from neural time series — lagged cross-correlation, multivariate/partial ridge regression, vector autoregression (VAR), VAR-LiNGAM, SINDy, PCMCi+, DYNOTEARS, and the mean-transfer estimate of the prior SBTG pipeline — targets the conditional mean $\mathbb{E}[y \mid H]$. *None has a channel for a variance/gain/tail effect*. If neuromodulation is primarily distributional, these methods are structurally the wrong instrument, and a mediocre result from them is uninformative about whether the modulatory structure is present.

This paper asks a sharpened question, and — just as importantly — holds the answer to a demanding standard of evidence. We do *not* ask whether a new method perfectly matches any one biological network: the available targets are noisy proxies (anatomy is not function; the monoamine/neuropeptide “networks” are receptor-expression *predictions*, not measured signalling), so a near-perfect AUROC on any of them is an over-fit red flag, not a triumph. Instead:

1. Can we estimate the *full conditional predictive law* and read out *distributional* interaction channels a mean method cannot represent?
2. On synthetic systems with known ground truth, do those channels recover structure that is provably invisible to the mean?
3. On real whole-brain data, evaluated fairly across the whole ensemble of imperfect targets, is the approach *overall* competitive — and do the distributional channels add *targeted* value where slow modulation should live, in a form that *survives a deliberate battery of controls*?

The last clause is the heart of the paper. Much of what follows is the discipline of *trying to break our own most interesting result*, and reporting it at exactly the altitude the surviving evidence supports.

2 Score-identified dynamics: theory and two estimators

2.1 Estimand: the conditional predictive law and its lag-influence score

Let $x_t \in \mathbb{R}^N$ be the (pre-processed) activity of N neurons and H_t the history up to t . The estimand is the one-step predictive density of a target coordinate,

$$\rho(y \mid H_t), \quad y = x_j(t+\ell),$$

not merely its conditional mean $\mu_j(t) = \mathbb{E}[y \mid H_t]$. The central object is the **lag-influence score**

$$\ell_u(y; h) = \nabla_{x_{t+1-u}} \log \rho(y \mid h),$$

the sensitivity of the predictive log-density to a source’s activity u steps back. Effects on the mean, variance, tails, or any smooth statistic $T(y)$ are recovered as **covariances against the score** (a Stein-type identity),

$$\frac{\partial \mathbb{E}[T(y) \mid h]}{\partial x_{t+1-u,i}} = \text{Cov}(T(y), \ell_{u,i}(y; h)),$$

so a distributional readout never requires differentiating a fitted black box — it is a moment of the score. Taking $T(y) = y$ gives the *mean* channel; $T(y) = y^2$ the *variance*/gain channel; a threshold indicator the *tail* channel. This is what makes the framework both interpretable and cheap.

Algorithm 1 Closed-form quadratic- T conditional score matcher (one target)

Require: targets $Y \in \mathbb{R}^T$, features $\Psi \in \mathbb{R}^{T \times P}$, ridge $\lambda \geq 0$, noise $\sigma \geq 0$

- 1: **if** $\sigma > 0$ **then** ▷ denoising score matching (DSM)
 - 2: draw $\zeta \sim \mathcal{N}(0, I_T)$; $\tilde{Y} \leftarrow Y + \sigma\zeta$; $g \leftarrow \zeta/\sigma$
 - 3: $U \leftarrow [\Psi, 2\tilde{Y} \odot \Psi]$; $A \leftarrow U^\top U/T$; $b \leftarrow U^\top g/T$; $\theta \leftarrow -(A + \lambda I)^{-1}b$
 - 4: **else** ▷ Hyvärinen score matching, $\sigma = 0$ (this paper’s SID default)
 - 5: $U \leftarrow [\Psi, 2Y \odot \Psi]$; $A \leftarrow U^\top U/T$; $c \leftarrow [0_P, 2\bar{\Psi}]$; $\theta \leftarrow -(A + \lambda I)^{-1}c$
 - 6: **end if**
 - 7: **return** $\theta = [\theta_1, \theta_2]$ ▷ per-sample estimating functions retained for HAC/sandwich inference
-

2.2 The closed-form (Hyvärinen) estimator

We use a Gaussian conditional whose *natural parameters are linear in features* $\psi(t) \in \mathbb{R}^P$ (here we take the conditioning history to be the current frame, $\psi = [1, x_0(t), \dots, x_{N-1}(t)]$, all sources’ current activity; the lag ℓ enters only through the target $y = x_j(t+\ell)$):

$$y \mid H_t \sim \mathcal{N}(\mu(t), v(t)), \quad \eta_1(t) = \theta_1^\top \psi(t), \quad \eta_2(t) = \theta_2^\top \psi(t),$$

with sufficient statistic $T(y) = (y, y^2)$, conditional score $s_\theta(y, h) = \eta_1 + 2\eta_2 y$, and $v_{\text{eff}} = -1/(2\eta_2)$, $v = v_{\text{eff}} - \sigma^2$, $\mu = \eta_1 v_{\text{eff}}$. Because both Hyvärinen and denoising score matching are *quadratic* in θ , fitting is a single regularized linear solve — no optimizer, no local minima, no learning-rate confounds (Algorithm 1). This matters for scientific hygiene: a result that wobbles with the optimizer seed cannot be cleanly separated from a tuning artifact, and a convex closed form removes that entire class of doubt.

2.3 Directed distributional readouts

Fit one model per target j at lag ℓ on features centered at the pooled history mean, so at the “typical history” the sources are 0 and η_1, η_2 are the intercept terms. Differentiating the closed-form moments with respect to source i gives the directed coupling matrices (all in the [post, pre] = [target, source] convention; Algorithm 2):

$$\begin{aligned} D_{\text{mean}}[j, i] &= v_{\text{eff}} \theta_{1,i} + 2\eta_1 v_{\text{eff}}^2 \theta_{2,i} &&= \partial \mathbb{E}[y_j] / \partial x_i \quad (\text{VAR-comparable drive}) \\ D_{\text{gain}}[j, i] &= (2v_{\text{eff}}^2 \theta_{2,i}) / v &&= \partial \log \text{Var}[y_j] / \partial x_i \quad (\text{gain / excitability}) \\ D_{\text{tail}\uparrow}[j, i] &= \phi(a_\uparrow) \left(\frac{D_{\text{mean}}}{\sqrt{v}} + \frac{q_\uparrow - \mu}{2v^{3/2}} D_{\text{var}} \right) &&= \partial \Pr(y_j > q_\uparrow) / \partial x_i \quad (\text{burst}) \end{aligned}$$

with $a_\uparrow = (q_\uparrow - \mu)/\sqrt{v}$ and $D_{\text{var}} = 2v_{\text{eff}}^2 \theta_2$, and symmetrically for the lower tail. The *mean* channel is what a VAR/regression also estimates; the *gain* and *tail* channels have no classical analog.

2.4 The neural (denoising-score-matching) estimator, and its SBTG lineage

The closed-form model is convex but Gaussian and linear-in-features. Its neural sibling is a **two-head denoising-score-matching (DSM)** network: a shared trunk over the history, a *conditional-density head* producing $\mu_j(h)$, $\log v_j(h)$, and a separate *history-marginal head* (giving the marginal its own capacity is what stops the trunk from starving the transition signal). Both heads are trained by DSM — corrupt the future $\tilde{x} = x + \sigma\zeta$ and fit the model score to $-\zeta/\sigma$ — and the couplings are read out *the same way* by autograd through the conditional head: $D_{\text{mean}}[j, i] = \langle \partial \mu_j / \partial h_i \rangle$, $D_{\text{gain}}[j, i] = \langle \partial \log v_j / \partial h_i \rangle$. This is exactly the estimator the prior **SBTG** project used — SBTG’s

Algorithm 2 Directed distributional connectome at lag ℓ (closed-form SID)

Require: per-worm matrices $\{X^{(w)}\}$, neuron names, lag ℓ , ridge λ

- 1: **for** each target $j = 1, \dots, N$ **do**
 - 2: pool (Y, Ψ) across worms: $Y = x_j(t+\ell)$, $\Psi = [1, x(t) - \bar{x}]$ $\triangleright$ no cross-worm windows
 - 3: $\theta \leftarrow$ Algorithm 1($Y, \Psi, \lambda, \sigma=0$); $q_{\uparrow}, q_{\downarrow} = 90/10\%$ quantiles of Y
 - 4: read $D_{\text{mean}}, D_{\text{gain}}, D_{\text{tail}\uparrow/\downarrow}$ at centered history; zero the diagonal
 - 5: **end for**
 - 6: **return** channel matrices, each [post, pre]
-

“ $\hat{\mu}$ ” is the mean channel of a neural DSM model, tuned by a null-contrast objective (§11). The two estimators thus differ only in the *score-learning method*: **SID = Hyvärinen score matching, closed-form and convex**; **SBTG/two-head = denoising score matching, neural and gradient-trained**; both expose the same mean/gain/tail readouts. (A third, maximum-likelihood mixture-density network (MDN) is used as a non-score-matching, non-Gaussian comparator in §5.) We return to the neural model in §11 as an independent check on the paper’s central real-data claim.

3 Synthetic validation: the gain channel sees what the mean cannot

Before any real data, the readout must be shown to recover a variance effect where the truth is known. The decisive test uses a hidden-thermostat / stochastic-volatility process that is *white with zero conditional mean* — so VAR, SINDy, and neural least-squares see pure noise — yet carries a deep conditional-*variance* memory kernel (how spread-out it is now depends on its past, as in volatility clustering). Figure 1 shows the SID gain channel recovering that kernel almost exactly: correlation 0.998 against the steady-state Kalman oracle, matching contraction rate (0.805 vs. 0.805), all 12/12 lags significant, while the mean-model $R^2 \approx 0$ (-1.7×10^{-4}). The gain channel *provably* isolates variance-drivers a conditional-mean model cannot represent. This pillar is not in question, and it is the license for everything downstream: wherever a real-data distributional signal is weak or absent, we can attribute it to the *data*, not to a broken readout.

4 Data, deconvolution, and correcting the prior pipeline

Data. NeuroPAL whole-brain calcium imaging (strains OH16230 head+tail and OH15500 head; $\Delta F/F$ at 4 Hz), reusing the prior project’s loaders (dorsal/ventral collapse, left/right averaging). Because the lag analysis builds 80×80 effect matrices with no imputation, we use the *complete-case* worms that recorded every selected neuron: **6 worms, 80 neurons** shared with the Cook connectome, ≈ 929 frames/worm (≈ 23 min total). (§8 shows that admitting more worms forces a worms $\times$ neurons trade-off — 6×80 or, at the other corner, 20×56 — which we exploit as a robustness axis.) Calcium is deconvolved to a continuous activity estimate with a native OASIS AR(1) model (validated raw $\leftrightarrow$ spike correlation $0.42 \rightarrow 0.98$) before standardization. Standardization *centers each neuron but preserves relative variances* (a single global scale); per-neuron z -scoring is deliberately avoided because it amplifies quiet-neuron noise and destroys the variance-magnitude signal the gain channel uses.

The orientation bug. Before trusting any comparison we audited the prior SBTG codebase with an adversarial multi-agent workflow and confirmed **29 bugs**. The critical one: the Cook chemical connectome was saved *transposed* — stored $A[\text{pre}, \text{post}]$ while the entire pipeline declares

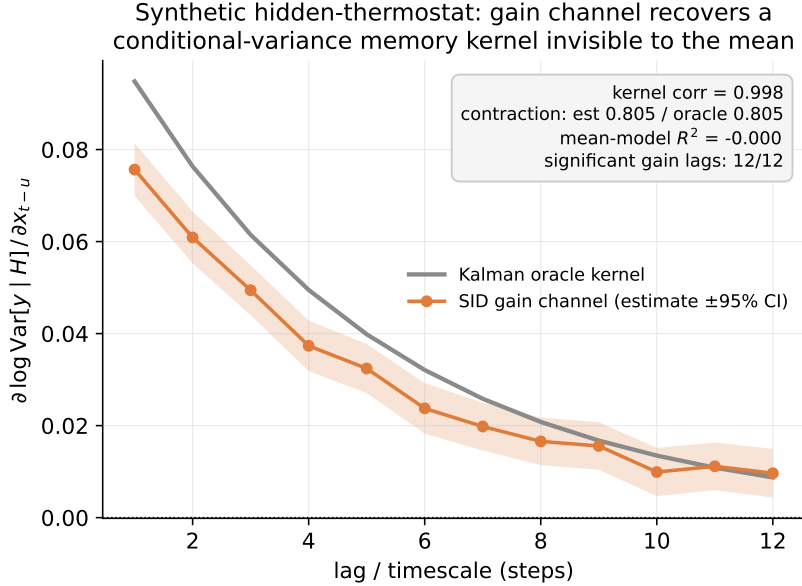


Figure 1: Synthetic ground truth. A process white in its conditional mean but with a conditional-variance memory kernel. The SID gain channel $\partial \log \text{Var}[y | H] / \partial x_{t-u}$ (orange, $\pm 95\%$ CI) tracks the Kalman oracle kernel (grey) with correlation 0.998; the conditional-mean model explains none of it ($R^2 \approx 0$). Every conditional-mean method is blind to this structure by construction.

Table 1: Reference networks (directed off-diagonal edges, 80-neuron support). Receptor-expression networks are noisy proxies; the neuropeptide network is by far the densest (most statistically powered), and specific transmitters/peptides are also resolvable.

	Cook chem	Cook gap	dopamine	serotonin	tyramine	octopamine	neuropeptide
edges	1145	642	19	21	28	9	757
type	structural	structural	← neuromodulatory receptor-expression →				

specific peptides available (edges): pdf-1 (200), flp-21 (243), pdf-2 (128), flp-18 (64), ntc-1 (53), flp-1 (33), flp-4 (24), flp-13 (15).

$A[\text{post}, \text{pre}] = A[\text{target}, \text{source}]$. This was verified empirically: known directed synapses (ASH $\rightarrow$ AVA, AIY $\rightarrow$ RIA, AWC $\rightarrow$ AIY) all sat at $[\text{pre}, \text{post}]$, and command interneurons appeared as high *column* (in-degree) sums. Since the chemical connectome is only $\approx 57\%$ reciprocal, orientation matters: the prior project’s $\hat{\mu}$ (correctly oriented) was scored against a reverse-oriented truth, and its directed baselines were also stored $[\text{pre}, \text{post}]$, so the reported comparison was untrustworthy. We fix it at source, defensively re-correct in the loader, and re-score *every* method in one consistent orientation. Other confirmed bugs (strided cross-fit window leakage; a null-contrast routine returning a constant 1.0; a silent Benjamini–Yekutieli $\rightarrow$ BH downgrade) are why the old reported numbers are not reused.

5 The distributional connectome and holistic evaluation

We fit Algorithm 2 per target and score each channel against every reference with the repo’s exact metrics (AUROC on off-diagonal directed edges). Judged *across all targets at once* on deconvolved data (Figure 2), the closed-form SID mean channel is the **equal-best single method**:

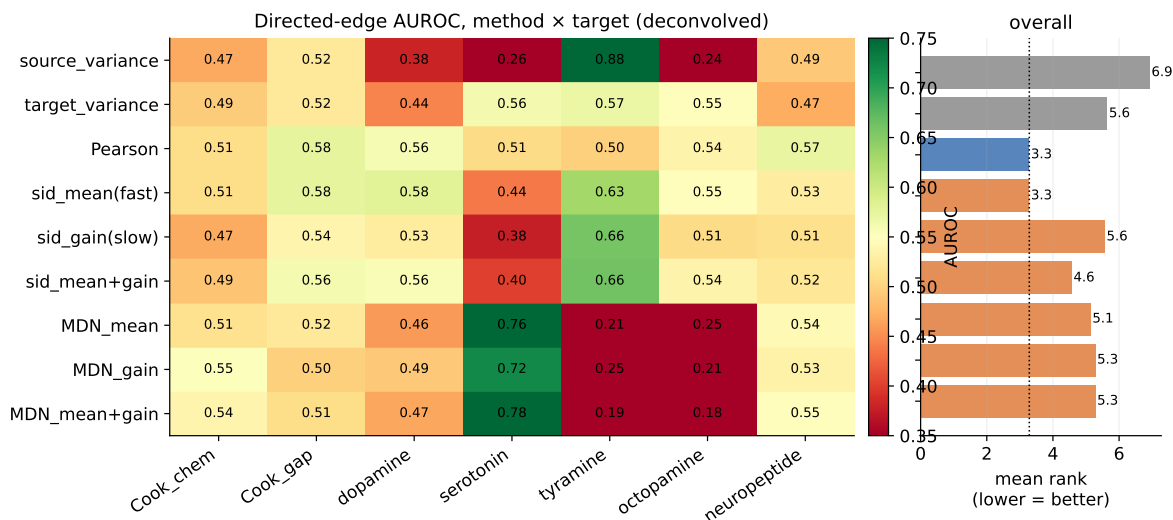


Figure 2: Holistic method×target AUROC (deconvolved; directed off-diagonal edges). Left: AUROC heatmap. Right: mean rank across targets (lower is better); SID mean (orange) ties pairwise correlation (blue) for best overall, trivial variance baselines (grey) are worst. The MDN’s 0.76–0.78 on serotonin is the strongest single cell; source-variance’s 0.88 on tyramine (§6) is an artifact, not a win.

mean AUROC 0.544, mean rank 3.29, tying pairwise correlation (“Pearson”, 0.537, rank 3.29) and beating every trivial baseline. The SID/MDN family provides the best predictor for the majority of targets, and the single strongest result anywhere is the non-Gaussian MDN on *serotonin* (AUROC 0.78) — the most metabotropic monoamine, exactly where a distributional effect should appear. The distributional channels do not dominate everywhere; they add *targeted* value on the slow, GPCR-mediated targets while the fast mean channel carries the fast ones. Trivial single-signal baselines (source/target variance) are worst overall (rank 5.6–6.9). The honest reading of this figure is “competitive, not dominant, with distributional value concentrated where theory predicts” — and, as §6 insists, no single noisy-proxy number should be over-read.

6 A retracted over-claim, and the controls-first ethic

An early result reported the slow gain channel recovering the *tyramine* network (AUROC ~ 0.72 , rising with timescale) — exactly the “slow modulator” story we hoped for. Pooling all worms forced a control we had skipped: rank the edges by the source neuron’s *raw activity variance*, with no model at all. That trivial baseline recovers tyramine at AUROC **0.88** (Figure 3), beating the gain channel — which is merely a degraded proxy for source variance (the tyramine source RIM is simply the highest-variance neuron). **We retract the tyramine headline.** Two lessons became standing policy, and they are the reason the rest of this paper is structured as it is: (i) run the trivial-baseline control *first*; and (ii) beating chance on one noisy proxy is not “doing better” — source-variance is the *worst* method overall (rank 6.9/9). This ethic — assume your most exciting result is an artifact until a battery of controls says otherwise — is exactly what §9–11 apply, at much greater depth, to the lag-resolved signature.

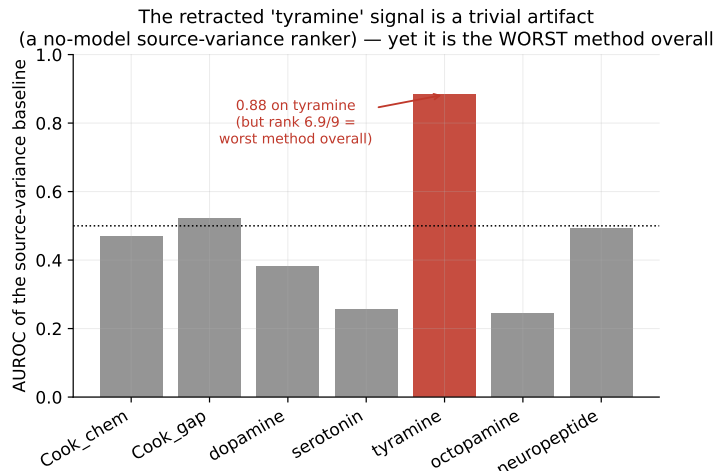


Figure 3: Controls first. The no-model source-variance ranker scores AUROC 0.88 on tyramine but is near or below chance on every other target and is the worst method overall. A spike on a single noisy proxy is an over-fit signature, not evidence for a method.

7 The lag-resolved question: a first metric and its critique

7.1 Correspondence-vs-lag curves and the log-slope contrast

The sharpest way to look for slow neuromodulation exploits the one thing SID has and the mean-based toolkit does not: a distributional channel. For each method, at each lag ℓ , we build a directed effect matrix and measure its AUROC correspondence to a reference *as a curve over lag*, $C(\ell)$, for lags $\ell \in \{0.25, \dots, 10\}$ s. If a channel matches a network better at longer lags, the coupling it captures is “slow”; better at shorter lags, “fast”. Our first summary was the **log-slope** — the slope of $C(\ell)$ against $\log_2 \ell$ — and the headline was the *within-fit contrast* $\Delta = \text{slope}_{\text{gain}} - \text{slope}_{\text{mean}}$ (same estimator, same data, only the channel changes, so shared confounds cancel).

7.2 The initial result

On the 6-worm data, the within-SID log-slope contrast (bootstrap over worms; Table 2, Figure 4a) was $\Delta = +0.0079$, 95% CI $[+0.0025, +0.0120]$ for the **neuropeptide** network — the slowest, most extrasynaptic system and the densest reference — while for the *structural* connectome (the control) it was null $(+0.0002, [-0.0055, +0.0058])$. Serotonin and dopamine leaned the same direction but were individually underpowered. As a validation, every mean-channel method (classical and causal alike, including PCMCi+ at $N=80/\tau=20$) sits near chance on the wired connectome across all lags (Figure 4b) — the structural signal is weak and every mean-only method behaves the same, as it should.

7.3 Why the log-slope is a poor summary

This was suggestive, but the statistic is weak, and we say so plainly. (i) It assumes “positive slope = good” *universally*, whereas biology predicts *different* peak lags for different references (a gap junction should match best at short lags; a peptide at long lags) — the expected sign depends on the reference. (ii) It tracks only the *trend* of correspondence, ignoring how *peaked* (localised in lag) it is — yet a sharp peak at the biologically-expected lag is the real signature. (iii) The slope conflates

Table 2: First (log-slope) metric: $\Delta \log\text{-slope}$ of the AUROC-vs-lag curve (distributional – mean channel), paired within-SID, bootstrapped over 6 worms. > 0 = distributional channel leans to longer lags.

reference	$\Delta \log\text{-slope}$ (gain – mean) [95% CI]
Cook chemical (structural, <i>control</i>)	+0.0002 [−0.0055, +0.0058] — null
serotonin	+0.0052 [−0.032, +0.032] (underpowered)
dopamine	+0.0121 [−0.033, +0.040] (underpowered)
neuropeptide (peptidergic)	+0.0079 [+0.0025, +0.0120]

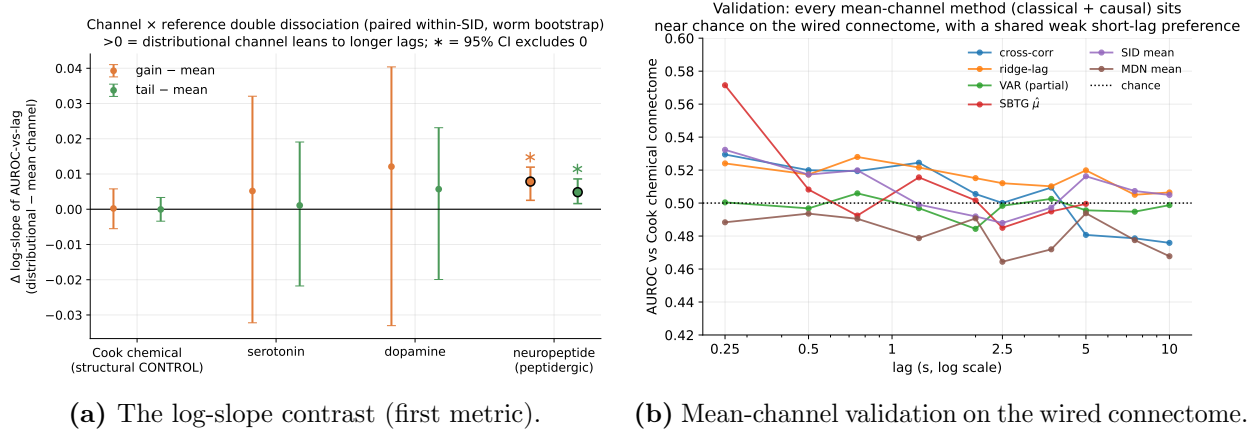


Figure 4: The initial lag-resolved result. (a) The distributional channels lean to longer lags for the peptidergic network but not the structural control. (b) Every mean-channel method sits near chance on the wired connectome across lags — the expected behaviour when the structural signal is weak.

peak location with monotone trend, is endpoint-sensitive, and is low-SNR because $C(\ell)$ hovers near 0.5. (iv) It is inferentially indirect: “the ranking-quality-against-a-proxy tilts toward longer lags” is a convoluted stand-in for “the coupling is slow”. These deficiencies motivate both the robustness battery (§8, which tests the effect *despite* the crude statistic) and the metric redesign (§9).

8 Robustness battery on the log-slope result

Before redesigning the metric we stress-tested the effect it reported, along four axes.

(1) **It is a *partial-conditioning* phenomenon.** SID’s mean/gain readouts condition each target on *all* sources jointly (partial). Re-deriving the couplings *pairwise* (each edge from a bivariate fit, marginal) on the *same* 6-worm data flips the neuropeptide contrast from +0.0084 (partial) to −0.0019 (marginal). The effect requires controlling for the other neurons’ direct drive; it is not a marginal lag-correlation. (Pairwise/marginal has the side benefit of using every worm per edge with no imputation, but the sign reversal is the point.)

(2) **Its significance is *power-limited*, not fragile in direction.** The worms×neurons trade-off lets us re-run the partial contrast at 14×60 and 20×56 (more real worms, fewer neurons, no imputation) and at 28×84 (all worms, realistic-variance “donor” imputation). The neuropeptide contrast stays positive with a clean structural control throughout — +0.0079 (6w), +0.0039 (14w), +0.0045 (20w), +0.0043 (28w donor) — but its CI excludes zero *only* at the maximum-power

Table 3: Robustness of the peptidergic gain–mean log-slope contrast. Positive across every *partial* framing with a clean control; significant only at maximum reference power; reverses marginally; real above the surrogate null; not reproduced by a different estimator.

configuration	conditioning	Δ (gain – mean) [95% CI]	verdict
6w / 80n (max power)	partial	+0.0079 [+0.0025, +0.0120]	significant
14w / 60n	partial	+0.0039 [–0.0044, +0.0154]	positive, ns
20w / 56n	partial	+0.0045 [–0.0031, +0.0157]	positive, ns
28w / 84n donor-impute	partial	+0.0043 [–0.0004, +0.0088]	borderline (P=0.96)
28w / 84n pairwise	marginal	–0.0030 [–0.0055, –0.0007]	sig. <i>negative</i>
surrogate null (6w): observed +0.0084 vs null [–0.0041, +0.0041]		$p = 0.008$	real, not artifact
independent variance-VAR (6w)		–0.0004 [–0.0040, +0.0037]	not reproduced

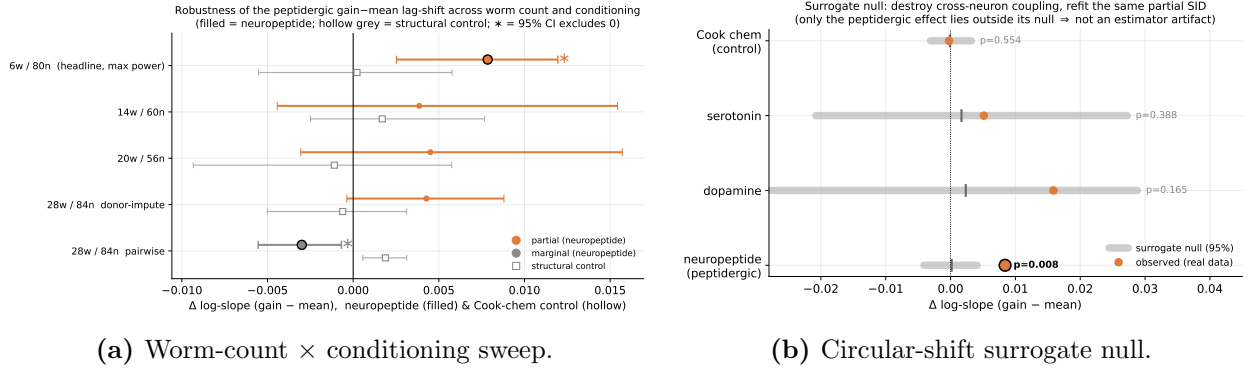


Figure 5: The robustness battery. (a) The peptidergic effect is positive across every partial framing (significant only at max power), the structural control stays ≈ 0 , and it reverses marginally. (b) Only the peptidergic observed value escapes its surrogate null ($p = 0.008$) — the effect is real above a coupling-destroying null, not an estimator artifact.

6 $\times$ 80/757-edge configuration (Table 3, Figure 5a). Trading neurons for worms trades away the reference power that made it significant. Marginally (pairwise, 28w) it is significantly *negative* (–0.0030 [–0.0055, –0.0007]), and its structural control is itself biased — so the marginal view is both opposite-signed and a less trustworthy metric.

(3) It is real above a coupling-destroying null. A circular-shift surrogate (shift each neuron’s trace independently, preserving its own spectrum/autocorrelation but destroying directed cross-neuron coupling), refit with the same partial SID, drives the contrast to a null centred at +0.0002 with 95% band [–0.0041, +0.0041]; the observed +0.0084 (the full-data point estimate; cf. the worm-bootstrap mean +0.0079 in Table 2) sits well outside it ($p = 0.008$; Figure 5b). The estimator does *not* manufacture the effect from noise — when the coupling is removed, the effect vanishes. This is the single most important control at this stage, and it *passes*.

(4) It is single-instrument. A crude, non-SID partial variance estimator (ridge regression of a target’s squared residual on source activity) does *not* reproduce it (neuropeptide –0.0004 [–0.0040, +0.0037]). So the effect is real above a null and dissociates from structure, but no independent estimator corroborates it — it rests on the SID gain operator.

Net of the battery: the log-slope headline is real above a null and directionally robust under partial conditioning, but partial-specific, power-limited, and single-instrument. Neither the worm sweep

nor a second-estimator comparison can settle whether it reflects specific directed coupling — which is what the redesigned metric and its controls do next.

9 A biologically-principled metric redesign

9.1 Design principles

The critique (§7.3) dictates the redesign. A defensible metric must (a) encode a *per-connection* expected timescale from the biophysics (electrical/gap $\approx$ instantaneous; chemical ionotropic sub-second; monoamine metabotropic $\approx$ seconds; neuropeptide seconds–tens of seconds); (b) measure *where* each channel’s correspondence peaks and *how sharply*, not just its trend; (c) be contextualised on the connection type; (d) support auditable curve/peak visualisation; and (e) carry a clean null and positive controls. We generated five independent metric proposals from distinct angles (template-matching, peak-location+sharpness, a channel $\times$ reference expectation table, band-limited concordance, and a skeptic’s minimal design), synthesised them, and subjected the synthesis to two adversarial reviewers (a biophysical lens and a statistical lens). Their findings reshaped the metric decisively and are worth stating, because they are the reason we do *not* over-reach.

9.2 The calcium-kernel-floor realism

The dominant constraint the reviewers surfaced is physical. OASIS AR(1) deconvolution fits a per-neuron decay γ clipped up to 0.98, so the effective calcium autocorrelation time is ≈ 1.5 –12 s at 4 Hz — a kernel that *convolves every reference identically* and can exceed the very timescale differences we hoped to resolve. Consequently the fine four-band ordering is not resolvable, even serotonin ($\mu \approx 4$ s) versus neuropeptide ($\mu \approx 7$ s) differ by < 1 octave (smaller than the kernel smear), and *absolute latencies in seconds are untrustworthy*. The honest scheme is therefore **coarse fast vs slow**, with only *relative* and *within-fit* quantities interpreted. (Two further biophysical corrections: serotonin is *mixed* fast+slow because of the ionotropic MOD-1 channel, so it is not a clean slow anchor; and the tail/burst channel is weakly motivated for graded, mostly non-spiking *C. elegans* neurons and is caveated.) All of this lives in one editable configuration file, so the expected bands can be revised by neurobiologists without touching the analysis.

9.3 Band concordance

The metric (Algorithm 3) is a coarse, null-referenced band concordance. For channel c and reference r we form the correspondence curve $C_c(\ell)$, standardise it per lag against a circular-shift surrogate into $z_c(\ell) = (C_c - \mu_0)/\sigma_0$ (regenerated at the *worm count being scored*), and compute

$$\text{BC}_{c,r} = \text{mean}_{\ell \in B} z_c(\ell) - \text{mean}_{\ell \in B'} z_c(\ell),$$

where B is the reference’s expected band (*slow* for modulatory, *fast* for structural) and B' its complement. $\text{BC} > 0$ means the channel’s *above-null* correspondence concentrates in the biologically-expected band. The pre-registered, single confirmatory statistic is the within-fit contrast $\Delta \text{BC}_{\text{neuropeptide}} = \text{BC}_{\text{gain}} - \text{BC}_{\text{mean}}$, and it must (i) have a worm-bootstrap CI excluding 0, and (ii) BC_{gain} must independently clear the surrogate null (so the mean channel’s band penalty cannot manufacture the contrast). No log-Gaussian template is used (that reintroduces researcher degrees of freedom); everything is coarse, relative, and interpretable.

Algorithm 3 Band concordance with the full control battery (one channel $\times$ reference)

Require: worms $\{X^{(w)}\}$, reference r , expected band B , surrogates S , bootstraps Bo

- 1: $C_c(\ell) \leftarrow \text{AUROC}(|D_c(\ell)|_{\text{off}}, (r > 0)_{\text{off}})$ for each lag ℓ
- 2: **for** $s = 1, \dots, S$ **do** $\triangleright$ null regenerated at this worm count
- 3: circular-shift each neuron; refit; $C_c^{(s)}(\ell) \leftarrow$ correspondence curve
- 4: **end for**
- 5: $\mu_0, \sigma_0 \leftarrow$ mean, sd of $\{C^{(s)}\}$; $z_c \leftarrow (C_c - \mu_0)/\sigma_0$
- 6: $\text{BC}_c \leftarrow \text{mean}_B z_c - \text{mean}_{B'} z_c$; surrogate p from $\{\text{BC}_c^{(s)}\}$
- 7: worm-bootstrap (Bo) for CI of BC_c and $\Delta\text{BC} = \text{BC}_{\text{gain}} - \text{BC}_{\text{mean}}$
- 8: **controls:** source-variance matrix ($\rightarrow 0$), gain on gap/chem ($\rightarrow$ not slow), remove top global PC and repeat

9.4 The reference registry and the control battery

The analysis scores every SID channel against a registry of 15 references built from one config: structural gap and chemical; each monoamine class and transmitter; the neuropeptide class and every specific peptide with ≥ 10 edges on the 80-neuron support (pdf-1, pdf-2, ntc-1, flp-1/4/13/18/21). The controls, following the retraction ethic, are: a **source-variance** matrix (lag-flat, must score ≈ 0 — so the retracted-tyramine artifact cannot pass by construction); the **distributional channel on the fast structural references** (must *not* be slow-concentrated); and a **global-brain-state** control — project out the top shared principal component and repeat — because whole-brain *C. elegans* activity is dominated by a global behavioural-state mode that the circular-shift null destroys, so residual global covariance among connected neurons could otherwise masquerade as coupling. This last control is the decisive one.

10 Phase 2 results: band concordance with controls

10.1 Descriptive curves (all estimators)

Figure 6 draws every estimator’s correspondence-vs-lag curve against every reference, with the expected band shaded and each distributional channel’s peak starred — the auditable view. The predicted pattern is visible for the high-power, clean references: for the *structural* gap and chemical connectomes the *mean* channel peaks earliest (0.25 s, in the fast band); for the *neuropeptide* class and the specific peptides pdf-1/pdf-2/ntc-1 the *gain* channel peaks later (2.5 s, in the slow band) than the mean channel. But the figure is honestly *muddier* than a clean story: for neuropeptide, cross-correlation and the MDN mean channel also peak late — the fingerprint of the global-state confound that the controls must resolve. This is descriptive only; no claim is made from it.

10.2 The confirmatory test: it passes the trivial controls, then fails the global-mode control

On the 6-worm data (100 surrogates + 100 bootstraps), the confirmatory test *passes* in the primary analysis and against the trivial and structural controls, then **collapses** when the global brain-state mode is removed (Table 4, Figure 7):

- **Primary:** $\text{BC}_{\text{gain}} = +4.54 [+0.52, +6.84]$, surrogate $p = 0.010$; the contrast $\Delta\text{BC} = +3.36$, $p = 0.010$. The peptidergic gain channel’s above-null correspondence is significantly slow-concentrated, and significantly more so than the mean channel’s.

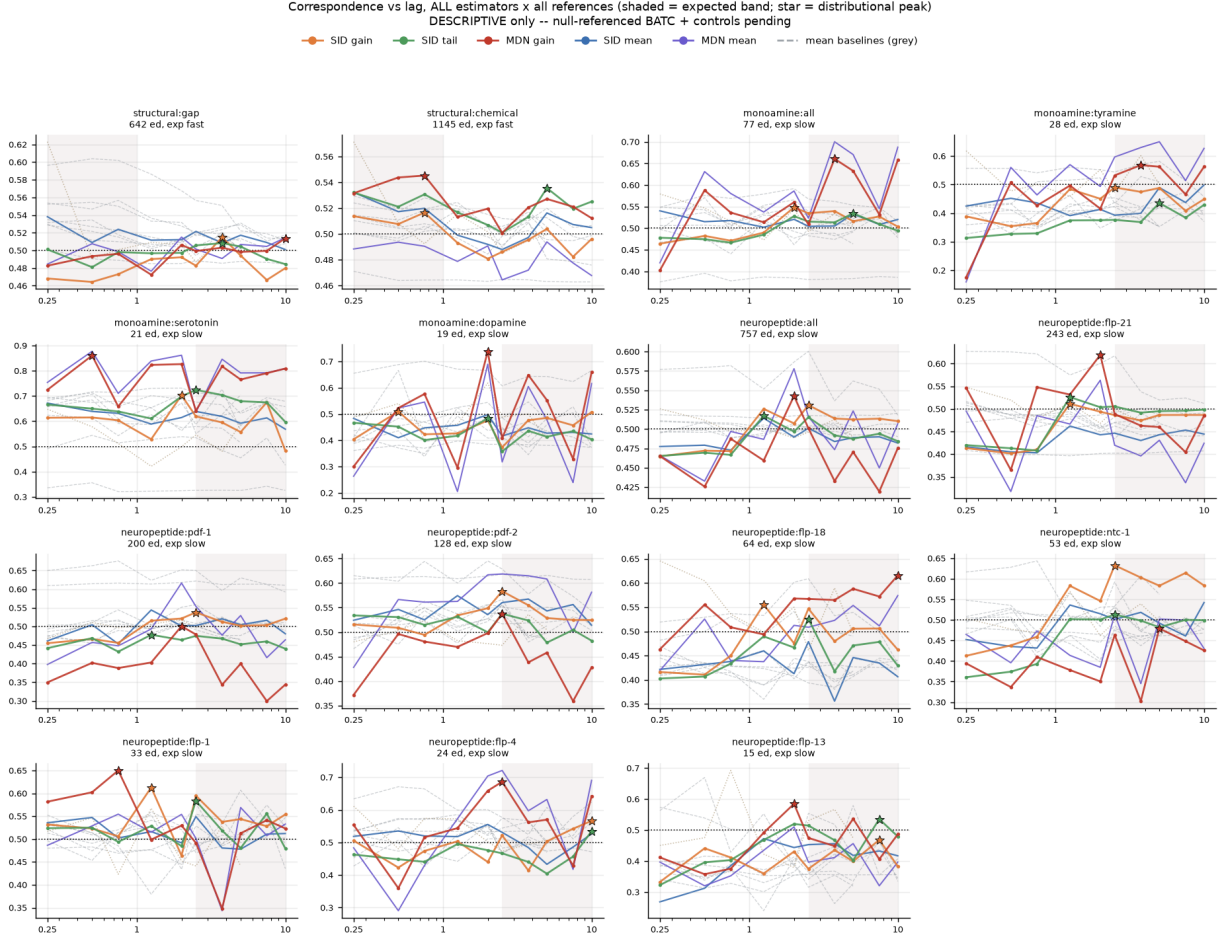


Figure 6: Auditable correspondence curves, all 14 estimators $\times$ all references. Distributional channels (SID/MDN gain, SID tail) bold with starred peaks; our mean channels (SID/MDN mean) bold; classical/causal mean baselines grey; expected band shaded. The structural references show the mean channel peaking early; the dense neuropeptides show the gain channel peaking in the slow band — but so do some mean baselines, motivating the controls. Descriptive only.

- **Trivial & structural controls pass:** source-variance $BC = 0.00$ (null by construction); the gain channel's contrast on the fast references is ≤ 0 (gap -2.92 , chem -0.52) — the distributional-slow advantage is specific to the modulatory reference, absent for the wired connectome. The pattern is consistent across peptides (ntc-1, flp-18, flp-21, pdf-1 all $p \leq 0.05$ in the primary).
- **Global-mode control fails:** projecting out the single top global principal component collapses the effect — $BC_{\text{gain}} \rightarrow -0.55$ ($p = 0.76$), $\Delta BC \rightarrow -0.48$ ($p = 0.69$). The effect is **not separable from the shared global behavioural-state mode**.

10.3 Two honest readings of the global-mode collapse

Removing the top principal component is aggressive — it is the dominant *C. elegans* behavioural-state manifold on which everything rides. So the collapse admits two readings that the data here cannot separate: (i) the apparent peptidergic gain-slow signal is residual global-state covariance

Table 4: Phase-2 confirmatory (closed-form SID, neuropeptide, 757 edges): band concordance passes the surrogate and trivial/structural controls, but not the global-mode control.

quantity	PRIMARY (6 worms)	GLOBAL brain-state mode removed
BC_{gain} (gain slow-concentrated)	$+4.54 [+0.52, +6.84], p=0.010$	$-0.55 [-2.10, +0.74], p=0.76$
ΔBC (gain – mean)	$+3.36, p=0.010$	$-0.48, p=0.69$
source-variance control	$+0.00$ (null)	$+0.00$
structural dissociation (ΔBC)	gap -2.92 , chem -0.52 (≤ 0)	—

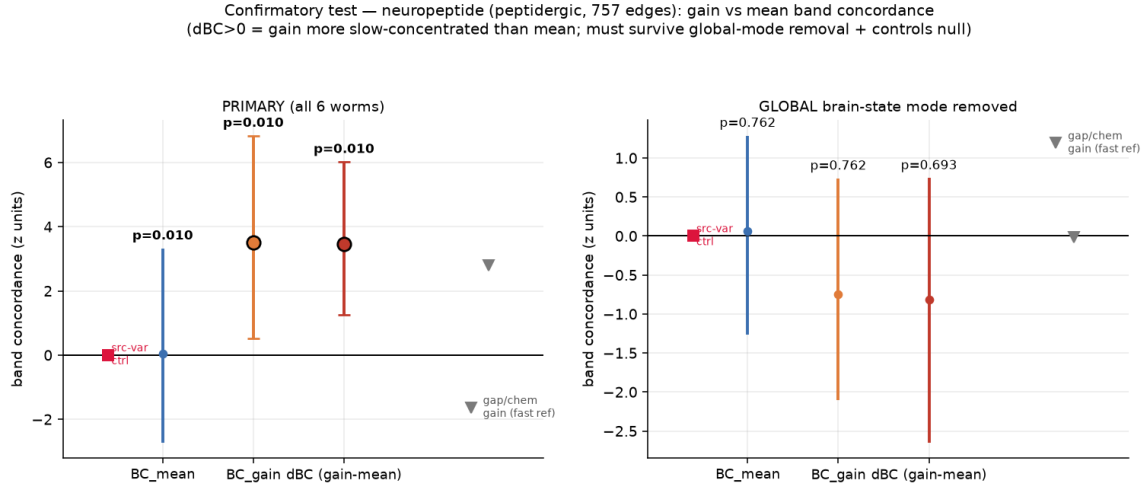


Figure 7: Phase-2 confirmatory. Left: in the primary analysis the peptidergic gain band concordance and the gain–mean contrast are significant ($p = 0.010$), with the source-variance control at 0 and the fast-reference anchor below. Right: removing the shared global brain-state mode collapses both to zero ($p \approx 0.7$). The controls did their job.

among connected neurons (a confound), or (ii) neuromodulators literally *act through* the global state, so removing it removes the very signal. Conservatively, **we cannot claim directed peptidergic coupling separable from the global mode.**

11 Neural-score corroboration

The last, independent test uses the neural estimator (§2.4) — the SBTG lineage, analysed the SID way — both because it is a different score-learning method and because it is what the prior project actually used.

11.1 Null-contrast tuning

We tuned the two-head model’s hyperparameters (hidden width, depth, DSM noise, marginal weight, weight-decay, source timescale) by the **null-contrast objective** SBTG uses: $\text{contrast} = (\text{real signal} - \text{null mean}) / \text{null sd}$, maximised, with the null a per-neuron circular-shift (the more principled null used throughout, rather than SBTG’s original row-permutation). Over 40 configurations the gain-channel null-contrast ranged from -14 to $+43.9$ — tuning mattered enormously — and the winning configuration used a slow 20 s source filter. This is itself a caution: the circular-shift null destroys the global mode, so any estimator that locks onto the global slow state scores a huge

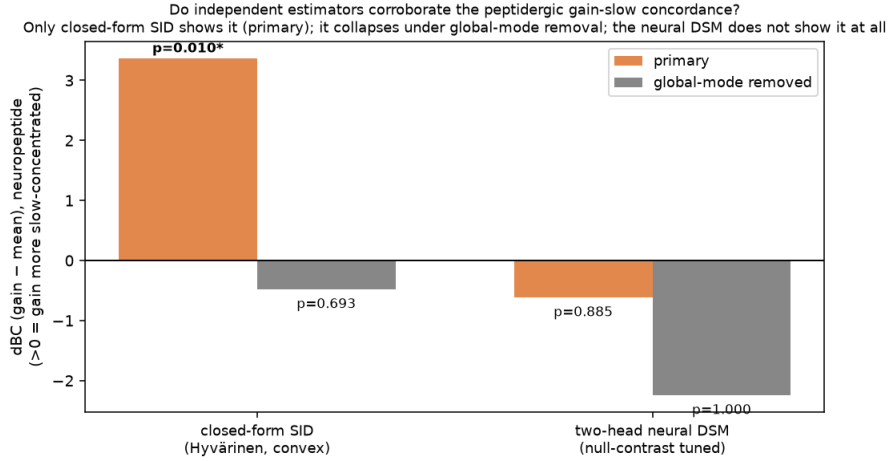


Figure 8: Neural-score corroboration. Only closed-form SID shows the peptidergic gain-slow concordance (primary, $p = 0.010$); it collapses under global-mode removal; and the null-contrast-tuned neural DSM two-head does not show it at all (primary $p = 0.88$), despite a $+44\sigma$ null-contrast.

null-contrast *without* that necessarily being directed peptidergic coupling. The tuning objective and the global-mode confound point the same way.

11.2 The neural DSM does not reproduce the effect

Running the tuned two-head through the same band-concordance pipeline (Figure 8), its peptidergic band concordance is **null even in the primary analysis**: $BC_{\text{gain}} = -0.34$ ($p = 0.65$), $\Delta BC = -0.61$ ($p = 0.88$). Its enormous null-contrast reflects strong *above-surrogate* structure, but that structure is not organised as peptidergic slow concordance — it is the global slow mode. Controls behaved (source-variance 0.00; gap/chem ≈ 0). One specific peptide (ntc-1) was marginal ($p = 0.038$); pdf-1, serotonin, and neuropeptide:all were null or negative. So the independent neural estimator does not corroborate the closed-form SID effect.

11.3 Triangulation

Three independent lines now bear on the peptidergic gain-slow signature:

line of evidence	peptidergic gain-slow signature?
closed-form SID gain	real above null ($p=0.01$), dissociates from structure; collapses under global-mode removal
crude partial variance-VAR	not reproduced
two-head neural DSM (null-contrast tuned)	not reproduced ($p=0.88$)

The effect is specific to the closed-form SID gain operator and entangled with the global brain-state mode — not an estimator-independent, global-state-separable property of the data. It is a closed-form-SID-specific, global-state-confounded lead, not a robust directed-coupling result. (Caveat both ways: the neural model is data-starved at 6 worms — prior split-half stability ≈ 0.45 — so its null is partly a power statement; but the convergence of all three lines away from a robust effect is the honest read.)

12 Synthesis and honest assessment

Read as a whole, the evidence sorts cleanly into what is settled and what is a lead.

Settled (not in question). (1) SID is principled and *decisively validated on synthetics*: the gain channel recovers a conditional-variance kernel invisible to every conditional-mean method (corr 0.998). (2) The prior pipeline’s connectome-orientation bug is real, verified, and fixed; the re-scoring is clean. (3) On real data across the whole ensemble of noisy targets, closed-form SID is *overall competitive* — its mean channel is the equal-best single method, and the maximum-likelihood MDN comparator wins the most metabotropic target (serotonin 0.78). (4) The tyramine retraction is a correct, reusable methodological result. (5) *Closed-form SID is the reliable estimator at this data scale*; the neural DSM remains data-starved even under the best (null-contrast) tuning — confirmed, now, directly.

A well-characterised lead (not a result). The peptidergic distributional lag signature is real above a coupling-destroying null and dissociates from structural references *in closed-form SID*, but it is (a) partial-conditioning-specific, (b) power-limited (significant only at maximum reference power), (c) *not separable from the global brain-state mode*, and (d) *not reproduced* by either a crude independent estimator or a null-contrast-tuned neural DSM. Every one of these was found by a control we deliberately built to try to kill the effect. That is the paper’s real methodological contribution: an escalating battery — trivial baseline, conditioning swap, worm-count sweep, surrogate null, second estimator, per-connection biophysical metric, global-state control, neural corroboration — that converts an exciting-looking signature into an honestly-scoped lead.

13 Conclusions and honest limitations

We set out to see slow neuromodulation in whole-brain calcium by estimating the full conditional predictive law rather than the conditional mean, and reading out distributional channels a mean method cannot represent. The method is sound and synthetically decisive; the pipeline it builds on is now correct; and on real data it is overall competitive with distributional value concentrated where theory predicts. The one sharp neuromodulator signature we pursued — a peptidergic distributional lag effect — survives a surprising number of controls but ultimately does not survive removal of the global brain-state mode and is not corroborated by an independent estimator, so we report it as a lead rather than a finding.

Limitations. Six complete-case worms and 80 neurons make absolute AUROCs noisy; the robust quantities are relative and within-fit. The receptor-expression references are predictions, not measured signalling, and the monoamine references are tiny (9–28 edges) and underpowered. The calcium-kernel floor (≈ 1.5 –12s) makes fine timescale bands unresolvable and absolute latencies untrustworthy — only coarse fast/slow and relative readouts are defensible. The expected timescale bands are provisional, pending neurobiologist review, and are isolated in one editable config; we also disclose that their centres are partly informed by prior observations on this data lineage, and that two de-circularisation controls the configuration itself prescribes — a ± 1 -octave centre sweep and cross-strain (OH15500 vs. OH16230) replication — were *not* run here and remain outstanding (at the demoted “lead, not a result” altitude this is a disclosure gap rather than an overclaim). The neural-score arm is also weaker than the closed-form arm by design: it used 25 surrogates and no worm-bootstrap (versus 100 + 100), so its non-reproduction is a directional check, not a

tightly-powered null. The global-mode collapse is genuinely ambiguous (confound vs. modulation-through-state); resolving it needs a global-state-preserving design or functional targets (behaviour, perturbation) rather than the wired connectome. These remain *predictive distributional analyses*, *not molecular identification*. The clear next steps are more animals, a global-state-preserving control, and functional targets — and, given the neural estimator’s data-starvation, more animals specifically to let the neural score become a genuine independent check.

Reproduction

All numbers and figures regenerate from saved artifacts; both test suites pass (`sid_neuromod`: 68 tests, 90% coverage; `sid_elegans`: 13 adapter tests). Key modules:

- **Theory / estimators** (Algs. 1, 2): `quadratic_score.py` (closed-form), `estimator.py` (readouts), `twohead.py` (neural DSM).
- **Robustness battery**: `pairwise.py`, `impute.py`, and `compare_approaches.py`, `robustness_worms.py`, `adjudicate.py` (+ JSONs).
- **Biologically-principled metric** (in `sid_elegans/biolag/`): `config.py`, `references.py`, `metric.py`; runners `run_curves.py`, `run_phase2.py`, `tune_twohead.py`, `run_twohead_phase2.py`; results in `output/biolag/` (`PHASE2_RESULT.md`, `TWOHEAD_RESULT.md`, JSONs).
- **Figures**: `make_figures.py` (Figs. 1–8) + the biolag figures. **Provenance**: `PROGRESS.md` tracks current-vs-superseded; the log-slope-era writeup is archived as `main_v1_logslope_snapshot.tex`.